

1.2 The main landforms associated with these processes

This chapter will explain:

- ★ the characteristics and formation of upland landforms
- ★ the characteristics and formation of lowland landforms
- ★ detailed specific example: the Rhône delta.

The characteristics and formation of upland landforms

The characteristic river landforms in upland areas ([Figures 1.18](#) and [1.19](#)) are a steep **V-shaped valley**, a steep gradient, interlocking spurs, potholes, waterfalls, rapids and gorges. In the upper course, much of the river's energy is needed to overcome friction. The rest is used to transport the load.



▲ **Figure 1.18** Meandering river in its upper course – a snow-covered mountain landscape in Switzerland

V-shaped valleys

Rivers in upland areas contain large boulders, which can erode the bed rapidly when the river is in flood. This results in the river cutting downward into its bed by vertical erosion to form steep V-shaped valleys. Soil and loose rock on the valley sides are washed down the steep slopes by overland flow into the river. This adds to the load.



▲ **Figure 1.19** The interlocking spurs of the Waitaki River, South Island, New Zealand

Interlocking spurs

Rivers begin to meander in the upper course. Erosion is concentrated on the outside banks of these small meanders. This eventually produces **interlocking spurs** that alternate on each side of the river. [Figure 1.19](#) shows that interlocking spurs are ridges of high land that project towards a river at right angles. They decrease in height towards the river. Interlocking spurs are eroded further down a river's course when lateral erosion takes over from vertical erosion as the dominant process in a river.

Potholes

Where the river bed is very uneven, pebbles carried by fast, swirling water can become temporarily trapped by obstacles. The swirling currents cause the pebbles to rotate in a circular movement, eroding (by abrasion) circular depressions known as **potholes** in the river bed ([Figure 1.20](#)).

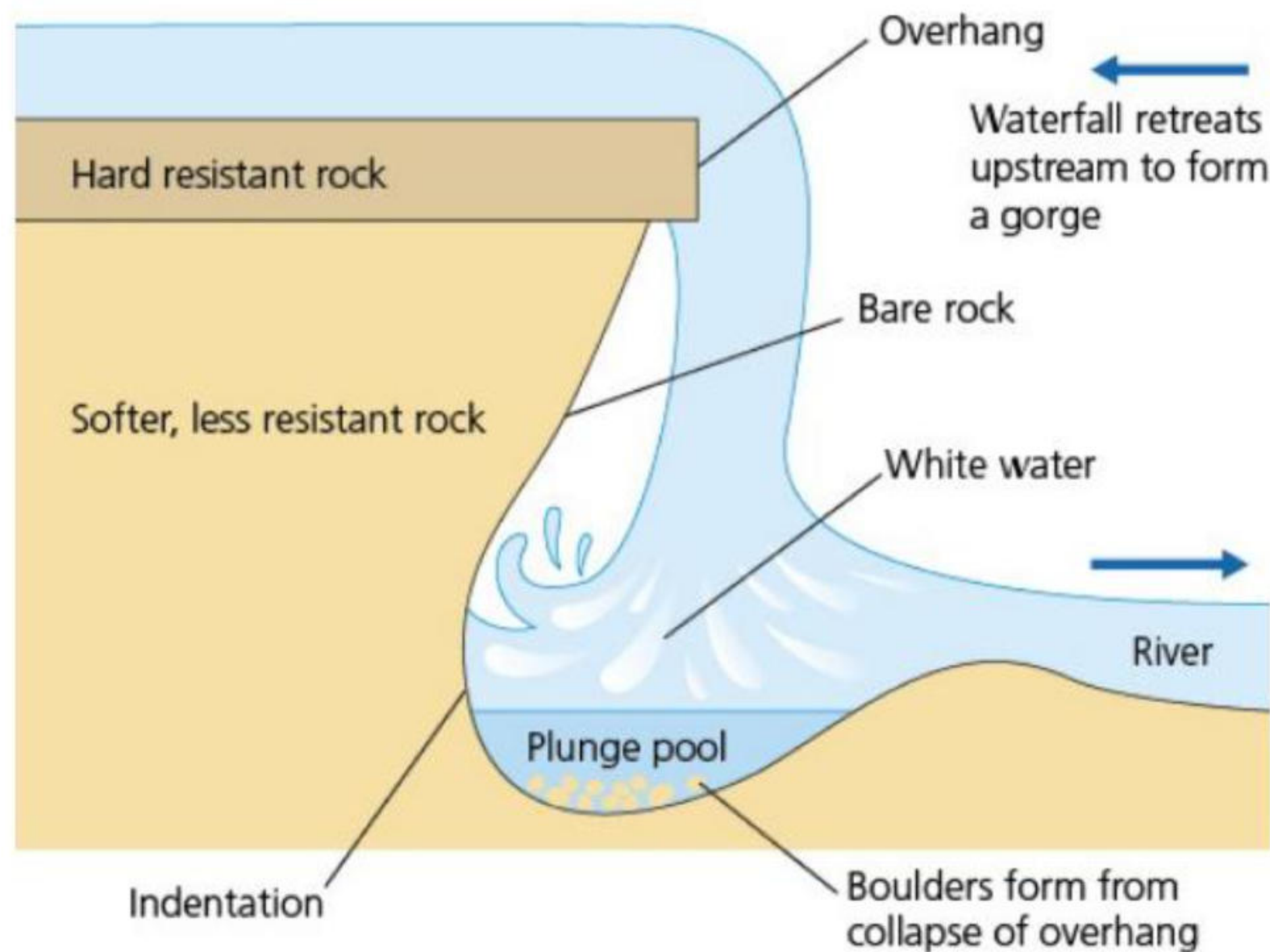


▲ **Figure 1.20** Potholes on the River Wharfe, Yorkshire, UK

Waterfalls and rapids

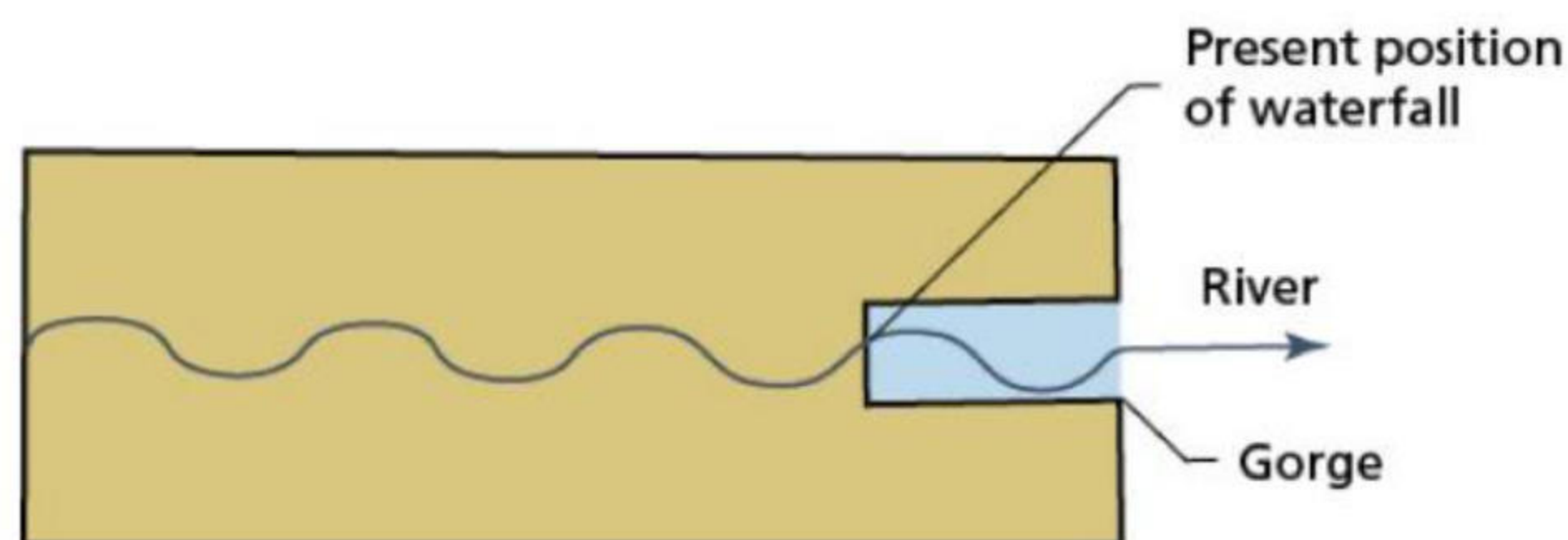
Waterfalls are the most spectacular feature of the upper course, but they can also be found in the middle course. They occur when:

- » there is a sudden change in the course of the river due to differences in rock hardness along the valley ([Figure 1.21](#))
- » a fault line has created an escarpment over which the river flows
- » glaciation has left a tributary valley hanging high above the main valley
- » a steep drop at the edge of a plateau has been formed by uplift of the land
- » a lava flow has crossed the path of a river, so it pours over its edge as a waterfall.



▲ **Figure 1.21** The formation of a waterfall

[Figure 1.21](#) shows what happens when a river flows from a band of hard, resistant rock on to a band of softer, less resistant rock. Waterfalls can form when the hard rock is horizontal, vertical or dipping upstream. The lower, softer rock is eroded more quickly, causing the hard rock to overhang. The undercutting is caused by corrosion and hydraulic action. The overhang steadily becomes larger until a critical point is reached. When this occurs, the overhang collapses. The rocks that crash down into the plunge pool are swirled around by the currents. This increases erosion, making the plunge pool deeper. The rocks in the plunge pool are eroded mainly by attrition.



▲ **Figure 1.22** The formation of a gorge of recession

This process, beginning with the collapse of a layer of hard rock, is repeated time after time. As a result, the waterfall retreats upstream ([Figure 1.22](#)), leaving behind a steep-sided **gorge**. The Niagara Falls ([Figure 1.23](#)) is a group of three waterfalls at the southern end of Niagara Gorge, which is 11 km long. The Falls span the border between the province of Ontario, Canada and the state of New York, USA.



▲ **Figure 1.23** The Niagara Falls, Ontario, Canada



▲ **Figure 1.24** Section of rapids along the River Wharfe, Yorkshire, UK

Sometimes very thin alternating bands of hard and soft rock cross the course of a river. This creates an uneven river bed and a zone of turbulent water known as **rapids** ([Figure 1.24](#)). Rapids can also form when a layer of hard rock dips gently downstream. The tallest waterfall in the world is the Angel Falls (979 m) located in the rainforest of Canaima National Park, Venezuela.

Activities

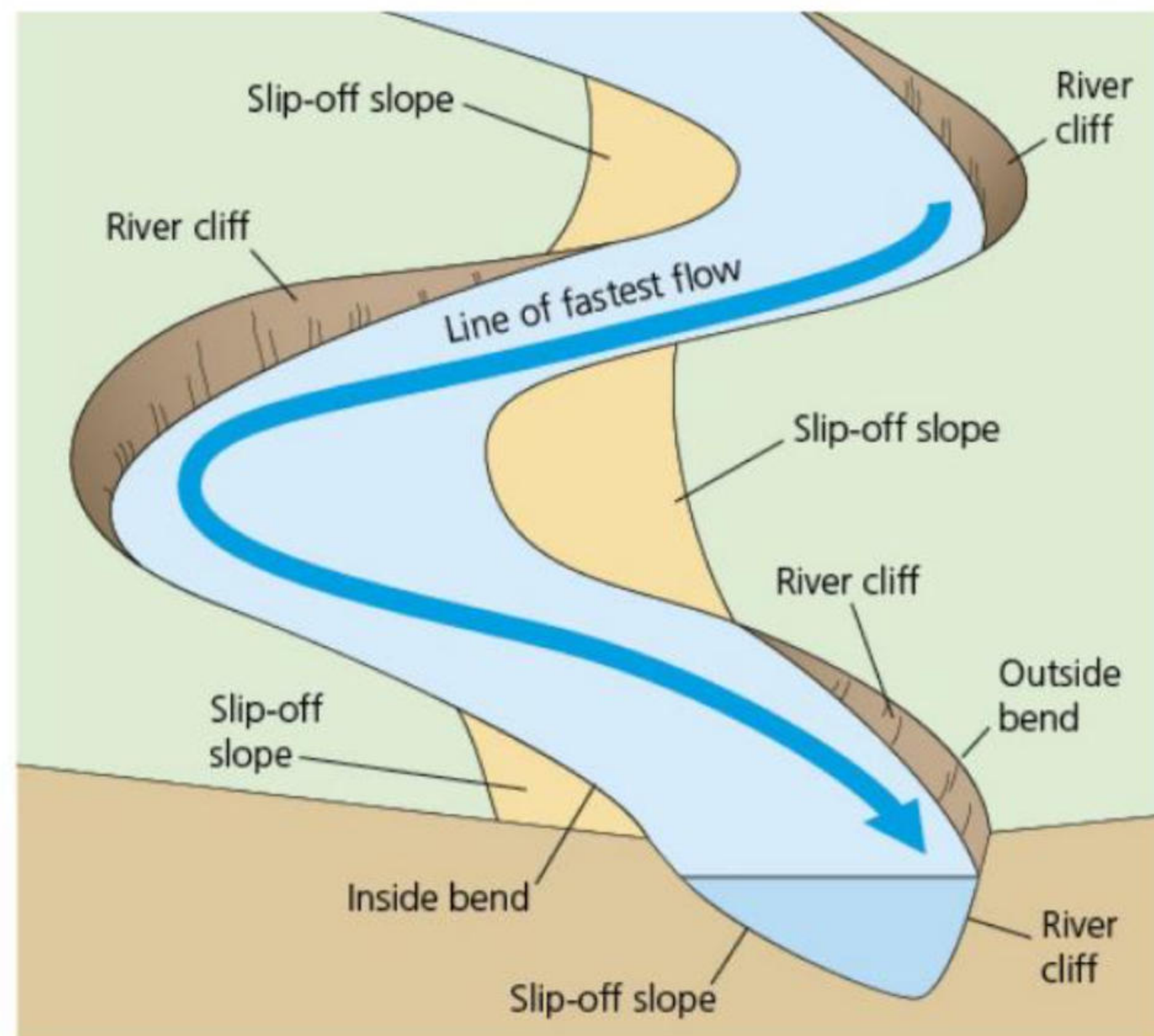
- 1 Which river landforms might you expect to find in an upland area?
 - 2 Explain the formation of interlocking spurs.
 - 3 Draw a fully labelled diagram to explain the formation of a waterfall.
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The characteristics and formation of lowland landforms

As more tributaries join the main river downstream, the volume of water increases. In lowland areas, if the rock is permeable, throughflow and groundwater flow also add water to the river.

Meanders and meander migration

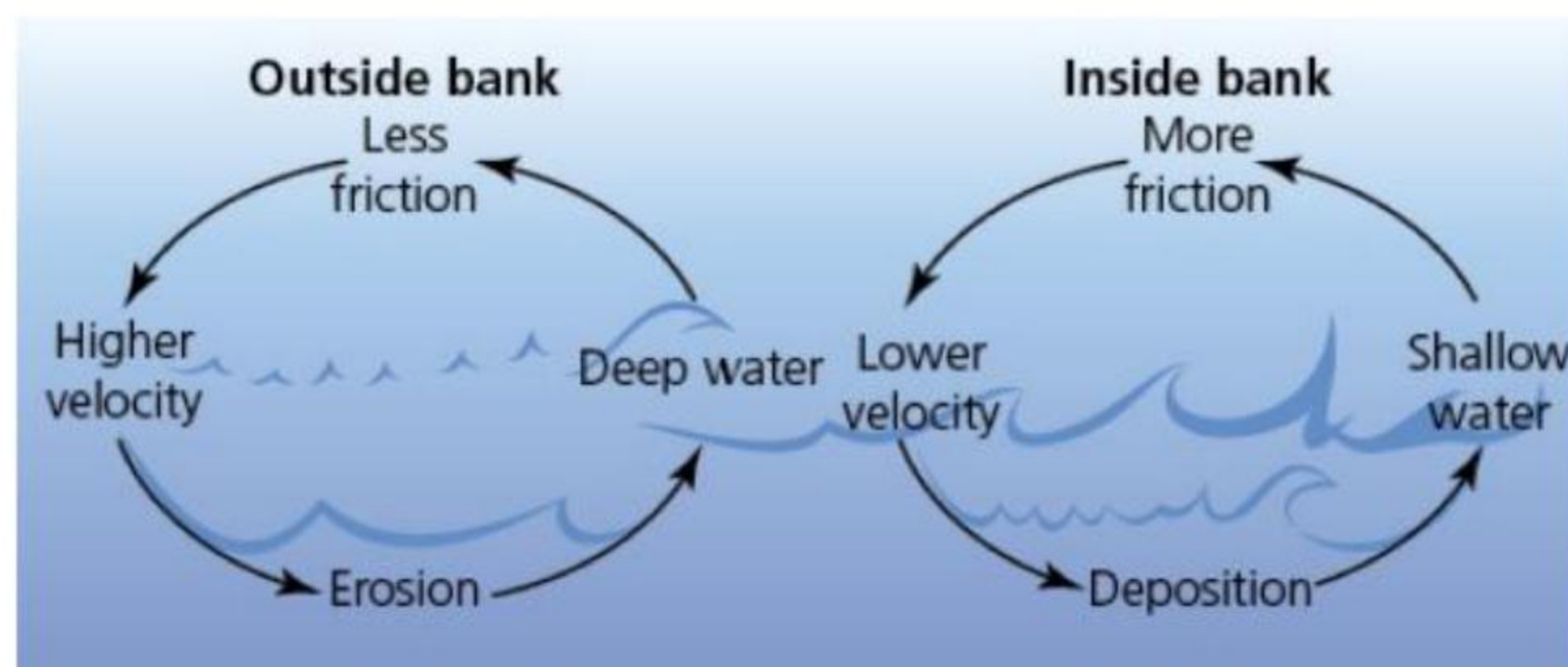
In lowland areas, lateral erosion takes over from vertical erosion as the most important process. As a result of this, meanders (wide bends in the river's course) become larger ([Figure 1.25](#)).



▲ **Figure 1.25** The cross-section of a meander

- » The current is fastest and most powerful on the outside bank of a meander, particularly on the downstream section. Erosion is relatively rapid. The outside bank is **undercut**. Again, the emphasis is on the downstream section. Eventually it collapses and retreats, causing the meander to spread further across the valley.
- » If the meander has already reached the side of the valley, erosion on the outside bend may create a very steep slope or **river cliff**.
- » The current on the inside of the meander is much slower. As the river slows, it drops some of its load and deposition occurs ([Figure 1.26](#)). This builds up to form a gently sloping **slip-off slope** ([Figure 1.27](#)).
- » This results in the water being shallow on the inside of the meander and deep on the outside.

Due to the power of lateral erosion, meanders slowly change their shape and position. As they erode sideways, they widen the valley. However, they also move or migrate downstream and erode the interlocking spurs, giving a much more open valley compared with the upper course.



▲ **Figure 1.26** The processes operating on the inside and outside banks of a meander



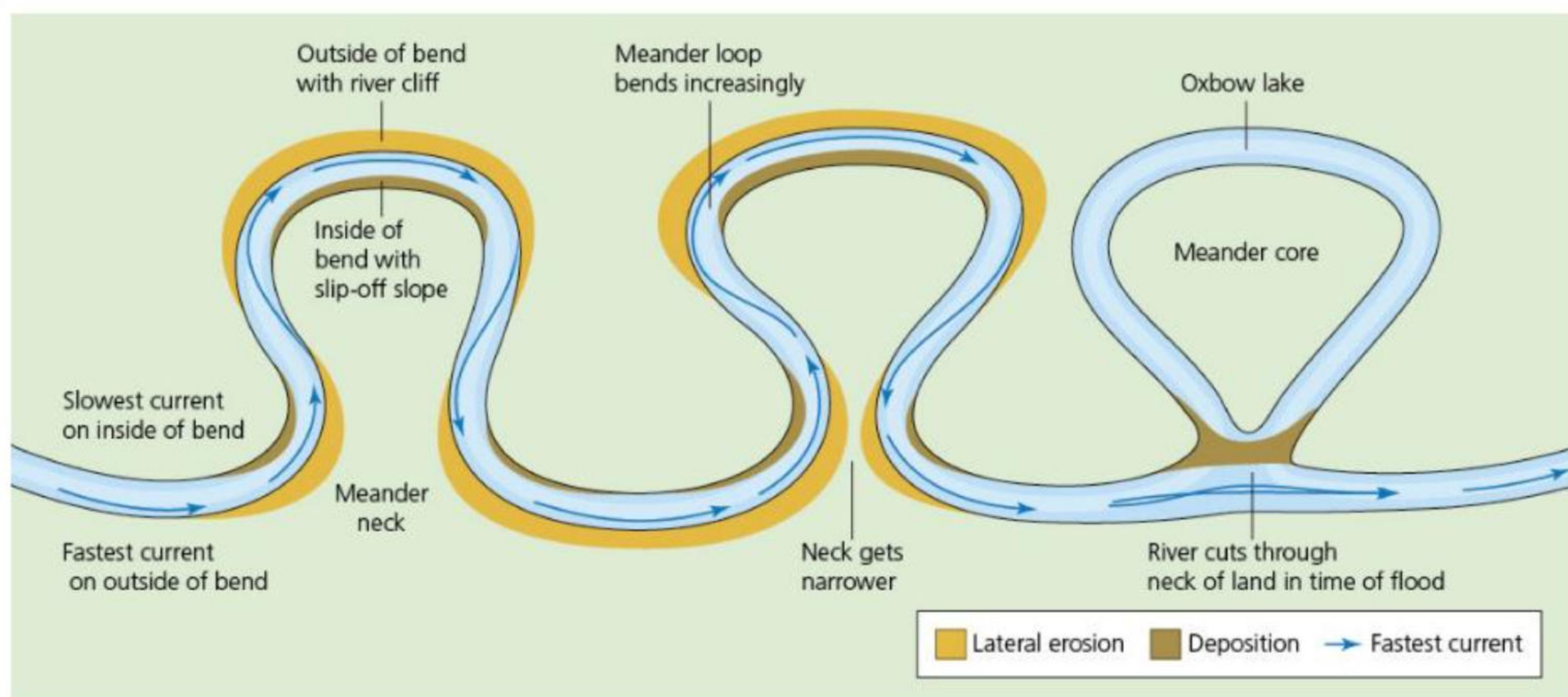
▲ **Figure 1.27** Meander and slip-off slope on the Marsyangdi River in the Annapurna Himalayan Range, Nepal

Oxbow lakes

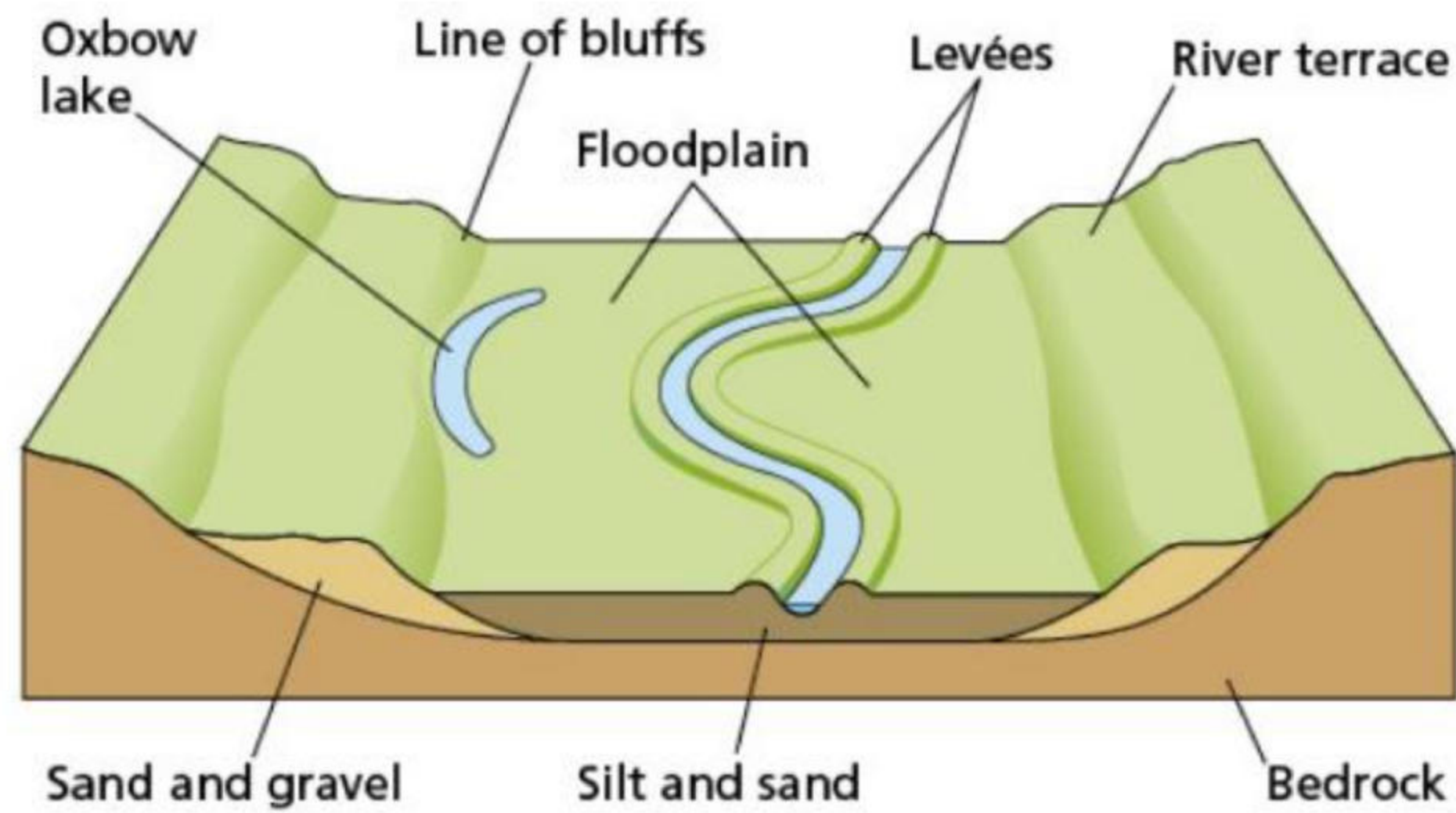
- » As a river flows towards its mouth, meanders become more pronounced and the valley becomes wider and flatter (see the left of [Figure 1.28](#)).
- » As erosion continues to cut into the outside bends of a meander, a **meander neck** may form (see the middle of [Figure 1.28](#)).
- » Eventually, when the river is in flood, it may cut right across the meander neck and shorten its course (see the right of [Figure 1.28](#)).
- » For a while water will flow both along the old meander route and along the new straight course. However, because the current will slow down at the entry and exit points of the meander, deposition will occur.
- » Over time, the meander will be cut off from the new straight course and leave an **oxbow lake**.

Floodplains and levées

A **floodplain** is a wide area of almost flat land on both sides of a river ([Figures 1.29](#) and [1.30](#)). It is formed by the movement of meanders, explained above. When discharge is high, the river is able to transport a large amount of material in suspension. At times of exceptionally high discharge, the river overflows its banks and floods the low-lying land around it. The increase in friction as the river water surges across the floodplain reduces velocity and causes the material carried in suspension to be deposited on the floodplain as alluvium. Alluvium is deposits of clay, silt and sand left by floodwater.



▲ **Figure 1.28** The formation of an oxbow lake



▲ **Figure 1.29** A cross-section of a river floodplain



▲ **Figure 1.30** The floodplain of the River Tuul, south of Ulaanbaatar, Mongolia

The heaviest or coarsest material will be dropped nearest to the river. This can form natural embankments alongside the river called **levées** (Figure 1.29). The lightest material will be carried towards the valley sides. Each time a flood occurs, a new layer of alluvium will be formed. This gradually builds up the height of the floodplain.

If the sea level falls and the river starts to cut down into its bed to adjust to the new coastline, the old floodplain will be left perched above the new river channel. This is then known as a **river terrace** (Figure 1.29).

Braided channels

Braiding is when a river divides for various distances into two or more channels (Figure 1.31). The channels are separated by islands of sediment. Braiding occurs when:

- » a river carries a very large load, particularly of sand and gravels, in relation to its velocity
- » the discharge changes rapidly from season to season and deposition occurs when the river current slackens.

The river then deposits so much sediment that the river channel becomes choked. The river is forced to split and find its way through its own deposits. The banks formed from sand and gravels are unstable. As a result, the channel becomes very wide in relation to its depth.

Interesting note

Seven of the world's rivers are more than 5000 km long – the Nile, Amazon, Yangtze, Mississippi-Missouri, Yenisei, Yellow and Ob.



▲ **Figure 1.31** Braided channel with vegetation established on a central island – Kyrgyzstan, Central Asia

Activities

- 1 What are the causes and consequences of meander migration?
- 2 With reference to [Figure 1.25 \(page 14\)](#), explain the processes operating on the inside and outside banks of a meander.
- 3 How are oxbow lakes formed?
- 4 Describe the formation of a floodplain.

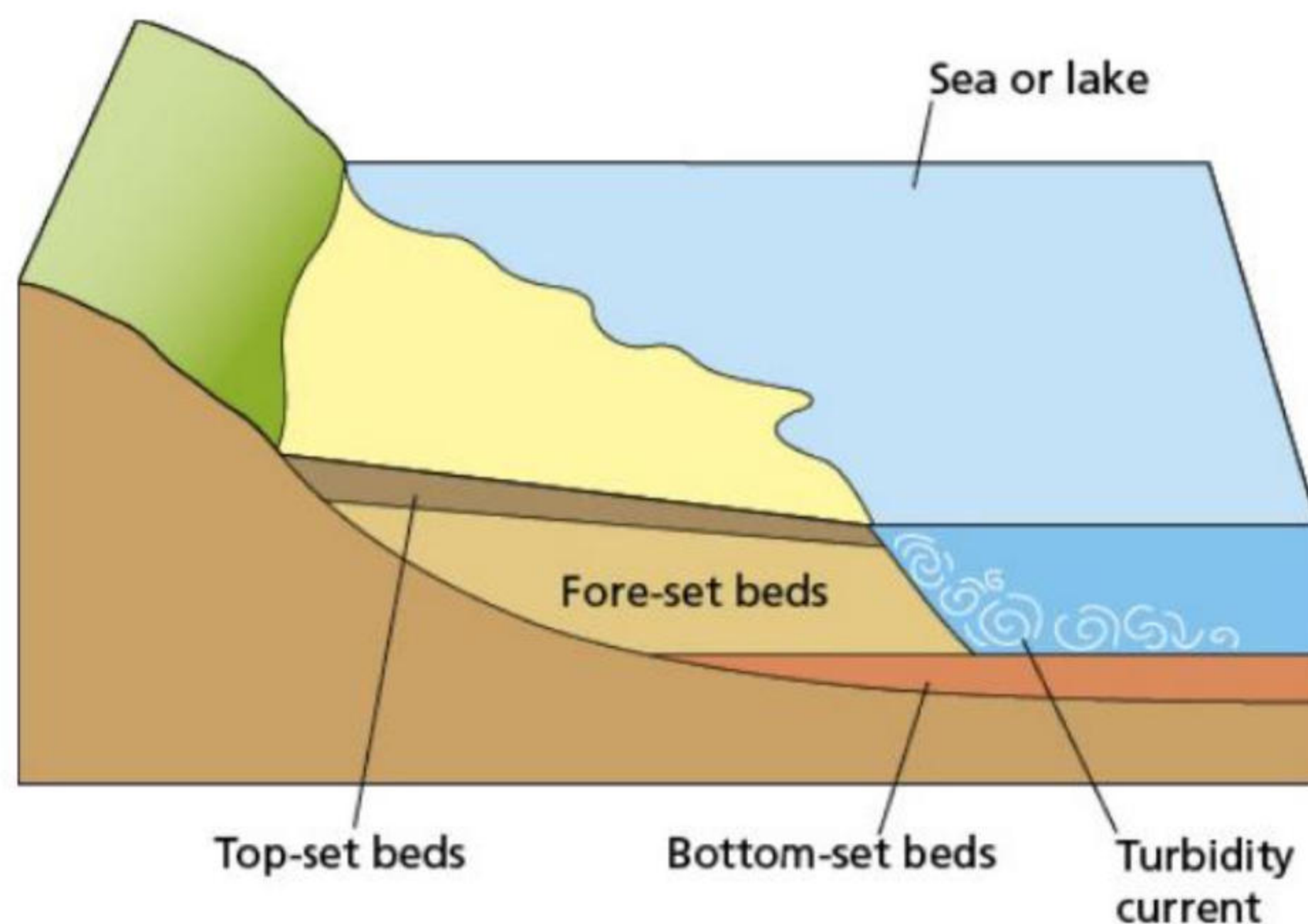
Deltas

A **delta** is a flat, low-lying deposit of sediment that is found at a river's mouth ([Figure 1.32](#)). For deltas to be formed, a river needs to:

- » carry a large volume of sediment – for example, rivers in semi-**arid** regions and in areas of intense human activity
- » enter a still body of water, which causes velocity to fall; the water loses its **capacity** and **competence**, hence deposition occurs, with the heaviest particles deposited first and the lightest last.

Deposition is increased if the water is salty, as this causes salt particles to group together, become heavier and be deposited. Vegetation also increases the rate of deposition by slowing down the water.

There are a number of stages in the formation of a delta. The first is the development of sandbanks in the original mouth of the river. This causes the river to divide, and then there is a period of repeated subdivision until there are a large number of distributaries flowing towards the sea. Each of the channels develops its own set of levées.



▲ **Figure 1.32** The formation of a delta

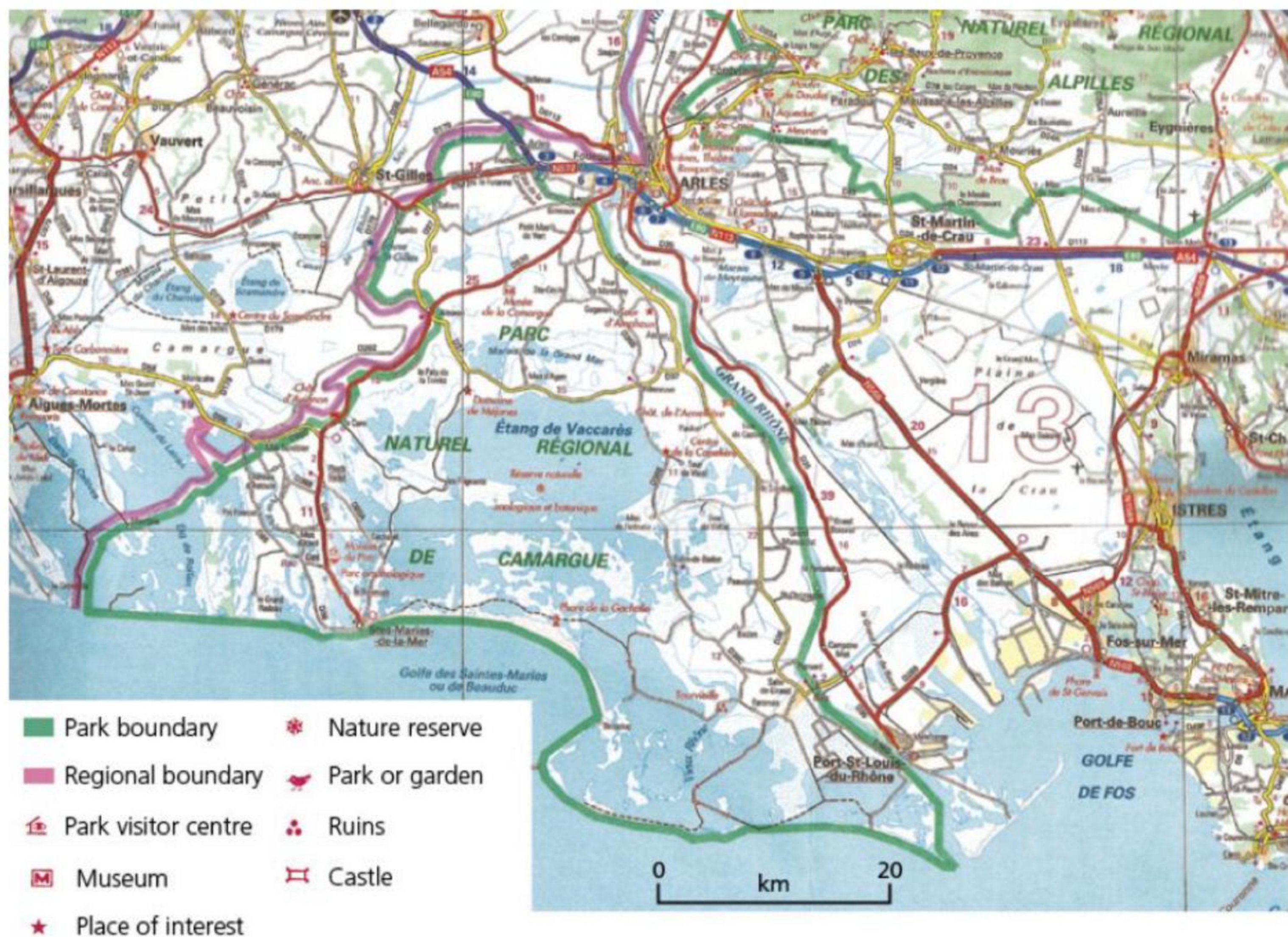
The Rhône delta

The Rhône river ([Figure 1.33](#)) divides into two main distributaries 4 km north of Arles. The east branch, the Grand Rhône, is the larger of the two, and carries 85 per cent of the Rhône's water into the Mediterranean.

- » At Arles the river is just 2 m above sea level and takes almost 50 km to reach the sea.
- » The delta is criss-crossed by numerous small islands, abandoned channels and active levées.
- » Most settlements and transport routes are located close to the river, where the land is slightly higher. Further away from the river, the land is lower, swampy and frequently covered with water.
- » Deposition by the river is estimated to be about 17 million m³ each year, or about 50 tonnes every minute. As the Mediterranean Sea has a very small tidal range, there are no currents to carry away these deposits.

Activities

- 1 Looking at [Figure 1.33](#), what is the length of the Grand Rhône between Arles and the sea?
- 2 How wide is the Rhône delta when it meets the sea?
- 3 What is the map evidence that:
 - a conservation
 - b tourismare important in the Rhône delta?



▲ **Figure 1.33** Map extract of the Rhône delta